

SUBGROUP CLAIM IDENTIFIED
"Treatment X works in Subgroup A but not B"

LEVEL 1: MINIMUM REQUIREMENTS
(All must pass to proceed)

1. PRE-SPECIFICATION

Was subgroup defined in protocol before seeing data?

? Unknown for Beta-blocker EF

2. BIOLOGICAL PLAUSIBILITY

Mechanistic rationale for differential effect?

X No clear mechanism for sharp threshold

3. INTERACTION TEST SIGNIFICANT

$p < 0.05$ for treatment $\times$ subgroup?

X $p = 0.069$ NON-SIGNIFICANT

Beta-Blocker EF Threshold:

Level 1: 0/3 passed X
Level 2: 0/3 passed X
Level 3: 0/2 performed X

TOTAL: 0/8 X

VERDICT: NOT VALIDATED

STOP if $p \geq 0.05$: INSUFFICIENT EVIDENCE

LEVEL 2: ROBUSTNESS CHECKS
(Assess reliability of finding)

4. ADEQUATE POWER

$\geq 80\%$ power to detect meaningful effect in subgroup?

X Beta-blocker: 40% power

5. FRAGILITY INDEX

FI > 5 for robust
FI > 10 for practice-changing

X Beta-blocker: FI = 3

6. OPTIMAL INFORMATION SIZE

Cumulative data reaches required information size?

X Below RIS

! ANY FAILED? $\rightarrow$ Interpret with CAUTION

LEVEL 3: VALIDATION (GOLD STANDARD)
(Required for practice-changing claims)

7. CROSS-VALIDATION

Effect holds in held-out data?

- Leave-one-trial-out
- Split sample

X Beta-blocker: Not performed

8. INDEPENDENT REPLICATION

Finding confirms in separate dataset?

- External data
- Prospective validation

X Not performed

FINAL DECISION

✓ **VALIDATED**

- All Level 1 passed
- Most Level 2 passed
- ≥ 1 Level 3 passed

$\rightarrow$ May inform guidelines

X **NOT VALIDATED**

- Failed ≥ 1 Level 1 test OR
- Failed Level 2 tests AND
- No Level 3 validation

$\rightarrow$ DO NOT use for guidelines